Lecture 7

Cheap Talk

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POLS W4732

Cheap Talk

(standard) signaling games: actions credibly reveal information because they are differentially costly for different types

Is it possible to communicate private information when actions' cost do not vary across types (i.e., signals are not costly)?

Lots of important settings have such "cheap talk"

- Congressional committees
- Client-lobbyist relations
- President and advisors
- Managers and employees in bureucratic organizations

If the sender and receiver do not agree on outcomes, sender may have incentive to distort information

A Cheap Talk Model

- Nature determines the state ω, drawn from a Uniform distribution on $[0, 1]$
- Sender (S) is informed of the state, Receiver (R)is not
- S submits a report, $r \in [0, 1]$
- R observes the report (not ω) and takes an action $y \in \mathbb{R}$

Payoffs

Cheap talk $\Rightarrow$ players' payoffs do not depend on report

Both S and R care about the action y, but they do not agree on the optimal y

$$u_S(y, \omega) = -(y - \omega - b)^2$$

$$u_R(y, \omega) = -(y - \omega)^2$$

Conditional on the state, S's most preferred a action is $\omega + b$ while R's is ω

b is S's (upward) "bias"

Exercise

Suppose that instead we had

$$u_S(y, \omega) = -(y - \omega - b)^2 - r$$

$$u_R(y, \omega) = -(y - \omega)^2$$

Would this be a standard signaling game?

Strategies

This is extensive form of incomplete information (like a signaling game)

Sender:

$$r(\omega) : [0, 1] \to [0, 1]$$

Receiver:

$$y(r) : [0, 1] \to \mathbb{R}$$

Use Perfect Bayesian Equilibrium

Can All the Information Be Transmitted?

Suppose S's strategy is $r(\omega) = \omega$

Consistency requires that R believe the state is ω whenever he sees the report ω

Sequential rationality implies that $y^*(r) = r$

Suppose we are in state ω'

- If S sends the message ω', he gets a payoff of
$$-(\omega' - (\omega' + b))^2 = -b^2 < 0$$
- If S sends the message $\omega' + b$ he gets a payoff of
$$-(\omega' + b - (\omega' + b))^2 = 0$$

There is no equilibrium in which $r(\omega)$ fully reveals the state

Can No Information Be Transmitted?

Suppose S adopts the strategy $r(\omega) = k$ for some constant k

Consistency requires that, upon observing a signal k, R's posteriors equal her priors

Thus, she will solve $\max_y \int_0^1 -(y - \tilde{\omega})^2 \, d\tilde{\omega}$

$$\int_0^1 -2(y^* - \tilde{\omega}) \, d\tilde{\omega} = 0$$

$$\iff -2y^* + 2\int_0^1 \tilde{\omega} \, d\tilde{\omega} = 0 \iff y^* = \frac{1}{2}$$

If R gets a report other than k then her beliefs are unconstrained. Suppose she keeps her priors, so $y^* = \frac{1}{2}$

Now there is clearly no profitable deviation for S

Can Some Information Be Transmitted?

Suppose S sends one of two reports, r_1 if $\omega < \omega_1$ (for some $\omega_1 > 0$) and r_2 otherwise

Upon seeing r_1, R must believe ω is uniform on $[0, \omega_1]$. Best response is

$$y^*(r_1) = \frac{\omega_1}{2}$$

Similarly, best response to r_2 is

$$y^*(r_2) = \frac{1 + \omega_1}{2}$$

Assume for any other report, R's beliefs are the same as under either r_1 or r_2

Partial Information (cont'd)

Given R's beliefs and best responses, need r_1 to be optimal when $\omega < \omega_1$ and r_2 to be optimal when $\omega > \omega_1$

For this to be true, we need S to be indifferent at state ω_1. Then, at lower states she will prefer lower action and at higher states she will prefer higher action

$$U_S(r_1|y^*, \omega_1) = -\left(\frac{\omega_1}{2} - (\omega_1 + b)\right)^2$$

$$U_S(r_2|y^*, \omega_1) = -\left(\frac{1+\omega_1}{2} - (\omega_1 + b)\right)^2$$

Indifference requires $\omega_1 + b$ to be midpoint btw $\frac{\omega_1}{2}$ and $\frac{1+\omega_1}{2}$

$$\omega_1 + b = \frac{\frac{\omega_1}{2} + \frac{1+\omega_1}{2}}{2} \iff \omega_1 = \frac{1}{2} - 2b$$

Partial Information (cont'd)

$$\omega_1 = \frac{1}{2} - 2b$$

- Since $\omega_1 > 0$, this is only possible if $b < \frac{1}{4}$. If bias is too large, no information can be transmitted.
- Suppose $b < \frac{1}{4}$.
- There is a PBE with two reports and some information transmitted
- This equilibrium Pareto dominates the "babbling equilibrium" with no information transmission (see next slide)

The Pareto dominance Argument

1. R must be better off, by Blackwell Theorem

$\Rightarrow$ she can ignore the report and always choose $1/2$:

$$-\int_0^1 (\frac{1}{2}-\omega)^2 d\omega < U_R = -\int_0^{\omega_1} (\frac{\omega_1}{2}-\omega)^2 d\frac{\omega}{\omega_1} - \int_{\omega_1}^1 (\frac{1+\omega_1}{2}-\omega)^2 d\frac{\omega}{1-\omega_1}$$

2. S must also be better off:

$\Rightarrow$ 2.a Expected action still $1/2$:

$$\omega_1 \frac{\omega_1}{2} + (1-\omega_1)\frac{1+\omega_1}{2} = \frac{1}{2}$$

$\Rightarrow$ 2.b we can write her expected payoff as

$$U_S = -\int_0^1 (Y_r(\omega)-\omega-b)^2 d\omega = -\int_0^1 [(Y_r(\omega)-\omega)^2 - 2b(Y_r(\omega)-\omega) + b^2] d\omega$$

$$= U_R - b^2 + 2b\int_0^1 [Y_r(\omega)-\omega] d\omega = U_R - b^2$$

where $Y_r(\omega) = y(r(\omega))$ is R's action as a function of ω

How Much Information Can Be Transmitted?

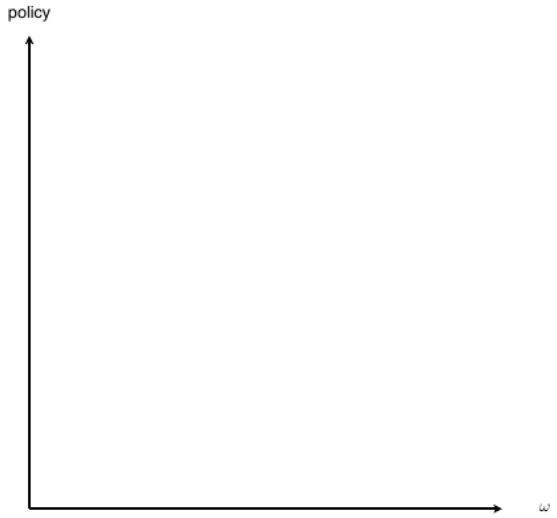
Suppose we want to allow an arbitrary number of messages K, each of which represents a partition of the space

If receive r_i choose $y^*(r_i) = \frac{r_i + r_{i-1}}{2}$, where $r_0 = 0$

At each state ω_i, S must be indifferent between $y^*(r_i)$ and $y^*(r_{i+1})$

$\omega_i + b$ must be the midpoint of $y^*(r_i) = \frac{r_i + r_{i-1}}{2}$ and $y^*(r_{i+1}) = \frac{r_{i+1} + r_i}{2}$

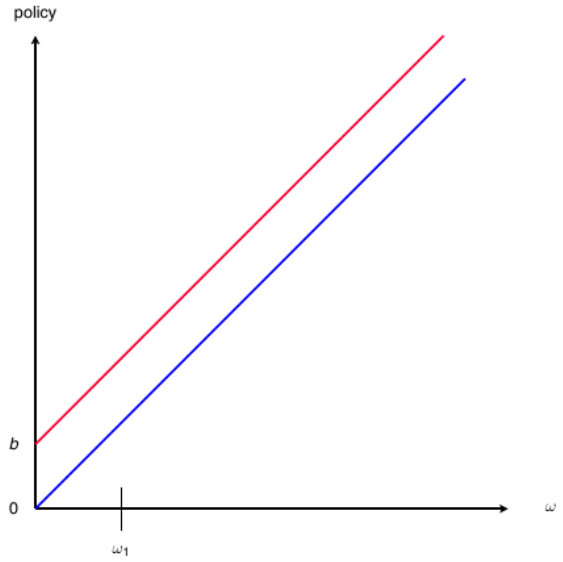
A Picture of the Construction



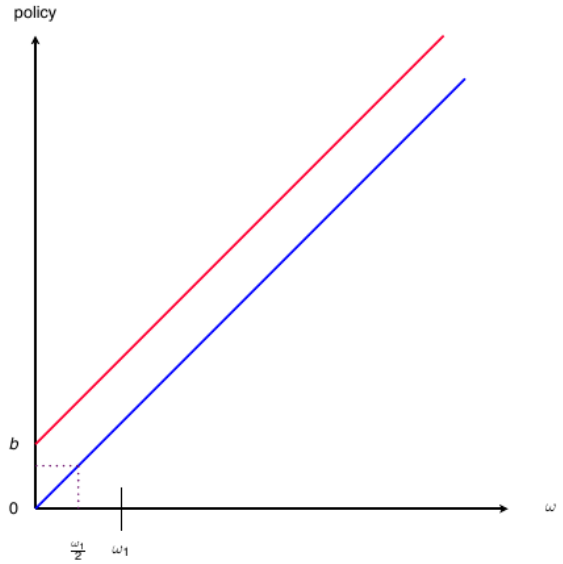
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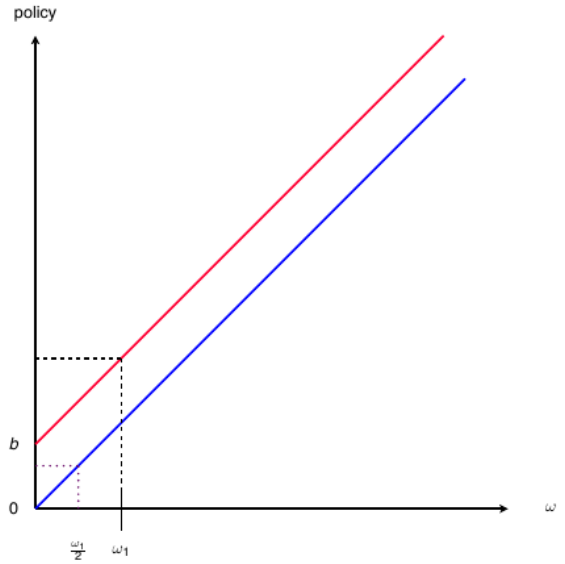
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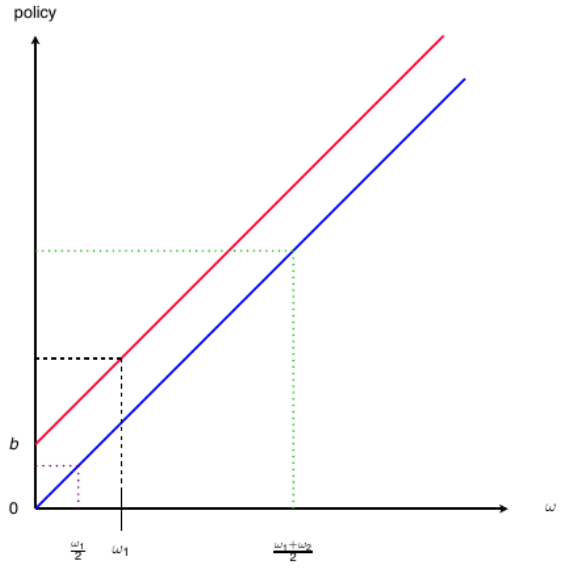
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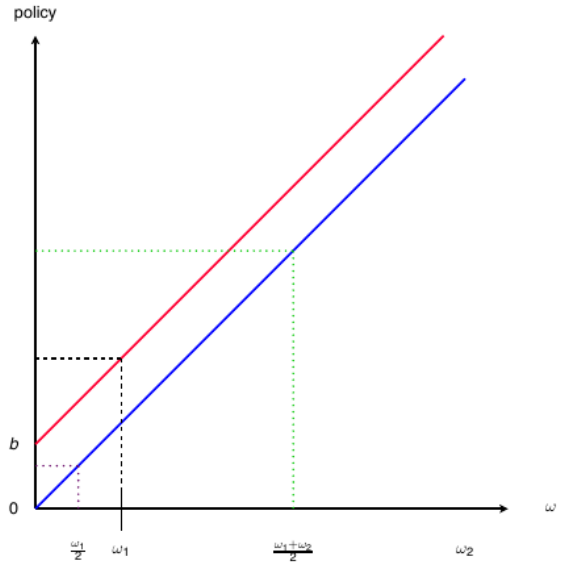
A Picture of the Construction



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A Picture of the Construction



How Much Information?

Notice in construction need

$$\omega_1 + b - \frac{\omega_1}{2} = \frac{\omega_1 + \omega_2}{2} - (\omega_1 + b) \iff \omega_2 - \omega_1 = \omega_1 + 4b$$

More generally, $\omega_{i+1} - \omega_i = \omega_i - \omega_{i-1} + 4b$

The total length of all intervals must be 1. So if there were K intervals, we would have[1]

$$K\omega_1 + 4b(1 + 2 + \ldots(K-1)) = 1 \iff K\omega_1 + 2bK(K-1) = 1$$

As b gets big, this becomes impossible. As b gets small, can have very fine-grained partitions

1. Recall the arithmetic series: $\sum_{i=1}^{n} = \frac{n(n+1)}{2}$.

Variations

- Multidimensional information $\Rightarrow$ full information transmission sometimes possible (Chakraborty and Harbaugh, 2007)
- Multiple senders $\Rightarrow$ full information transmission sometimes possible (Krishna and Morgan, 2001)
- Under verifiable information (message space depends on type: $M(t)$; e.g., because of no lying) can support full revelation (Grossman, 1981; Milgrom, 1981; Seidman and Winter, 1997)

Applications: Banks 1990

A model of agenda setting in which the informed sender proposes a policy to a receiver (decision-maker), who can either accept or reject the proposal

If rejected outcome is status quo, which is only known to the sender

Sender's strategy may convey information to the decision maker, since signaling is costly

However, since receiver has two actions, fully revealing equilibria need not exist

Refinements limit yield outcomes in which only one proposal is accepted in equilibrium w.p. 1, but typically a continuum of equilibrium outcomes

Applications: Matthews 1989 QJE

A cheap-talk model of veto threats; a Chooser (C), and a Proposer (P)

Preferences: distance between action (x) and ideal point t_j^*

P's ideal point is common knowledge (for simplicity, $t_P^* = 0$), t_C^* is drawn from $[0; \bar{t}]$

C learns her type, then sends a cheap-talk signal to P, who responds with a proposal, which can be accepted or rejected ($\rightarrow$ status quo)

Applications: Matthews 1989 QJE

There may be size-two (perfect Bayesian) equilibria in which two outcomes arise with positive probability (but no more)

In a size-two equilibrium

- P offers t_P^* to types of C whose message indicates that their ideal point is low, which is always accepted.
- If C indicates otherwise, P makes a compromise offer that is sometimes accepted and sometimes rejected